

CONVOLUTIONAL NEURAL NETWORKS

Pain points

How do we determine how many convolutional layers we require for a given problem and its parameters (tuple determining the size of each convolution)

In which scenario(s) would it be beneficial to have two or more convolutional layers in a row before pooling?

Is there a rule of thumb for 1) the dimensions of the local receptive field and 2) the number of neurons in the dense layer after flattening and before the output layer?

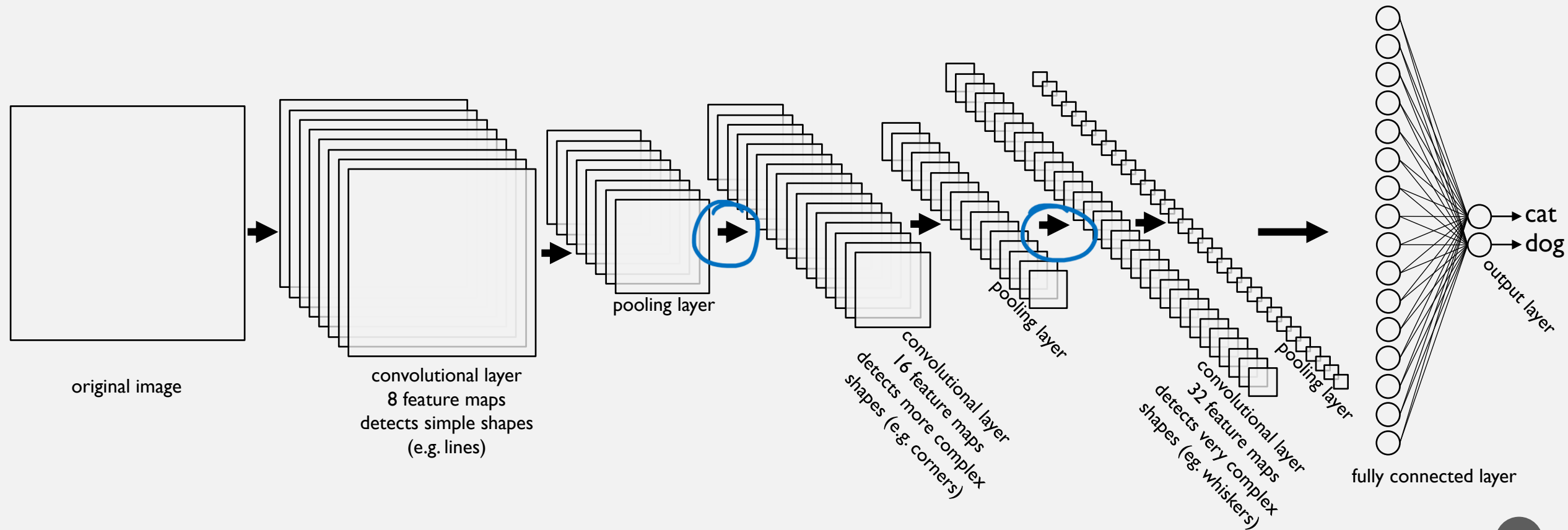
Start small & simple:

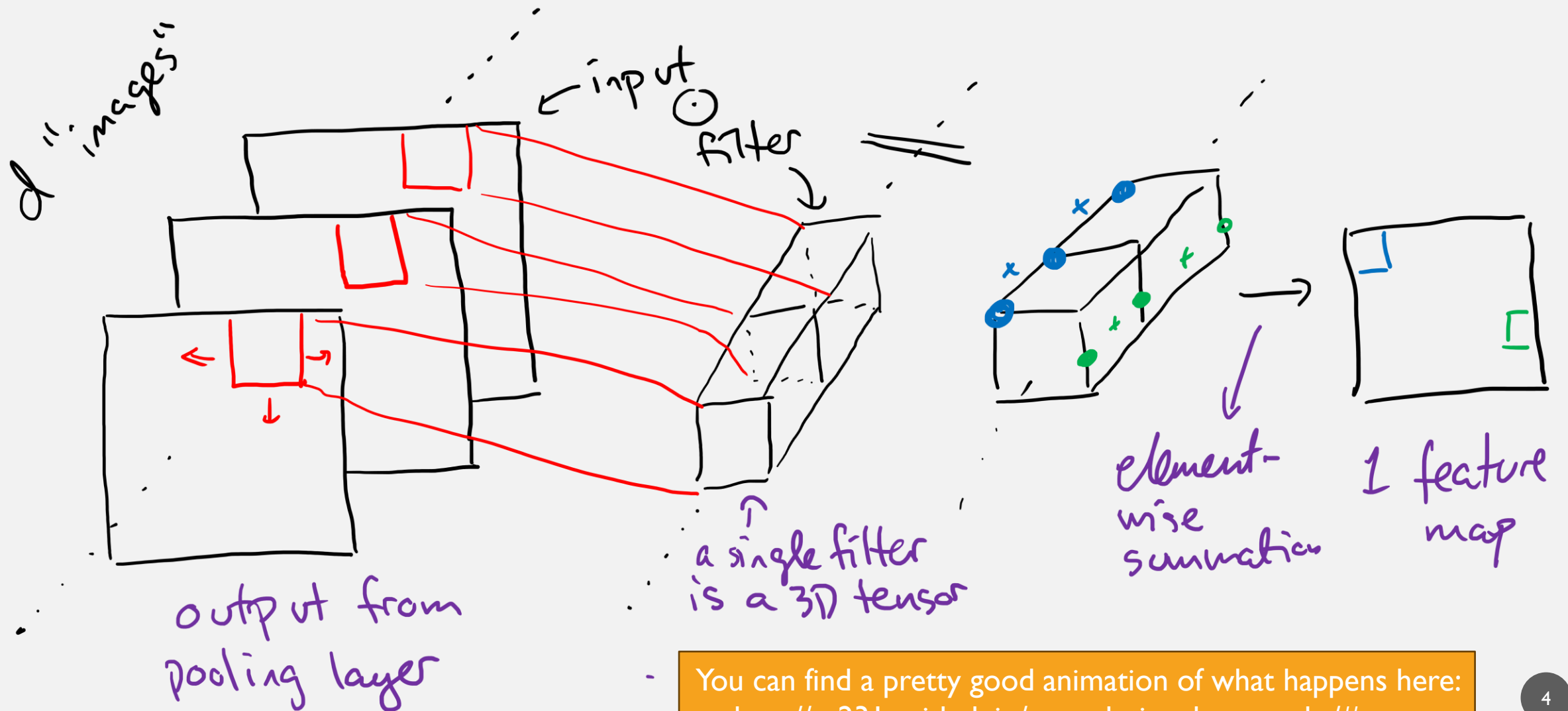
A few conv. layers, filters 3×3 or 5×5 ,
one FC layer, $n_{\text{neurons in FC}} \gg n_{\text{output}}$

Increase complexity (# of convolutional layers, # of feature maps, filter size, # of FC layers, # of neurons in FC layers) as needed to improve accuracy but not so much that you start overfitting.

cnn

What happens from a pooling layer to a convolutional layer? Like, how are they connected?





You can find a pretty good animation of what happens here:
<http://cs231n.github.io/convolutional-networks/#conv>